

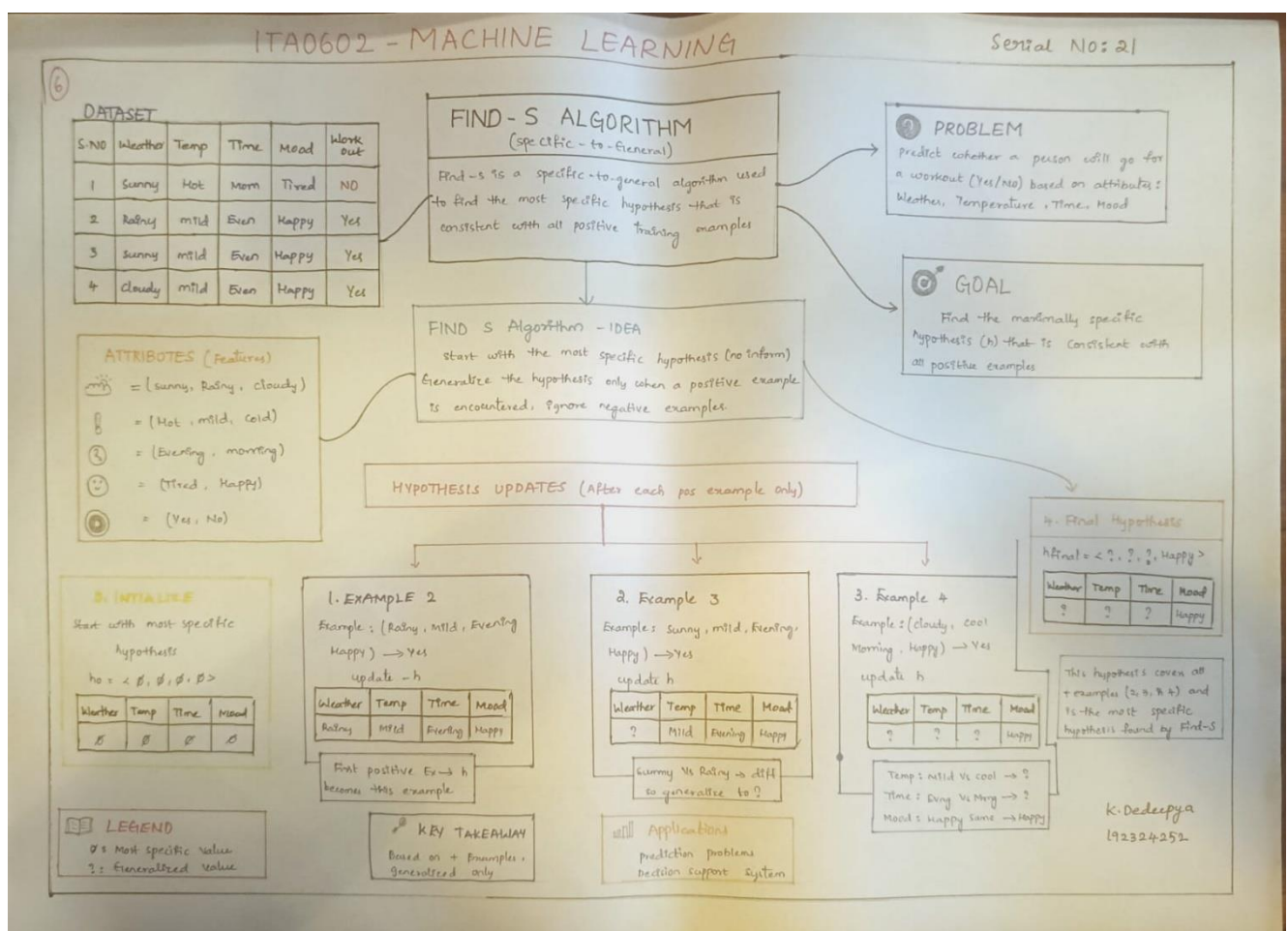
ITA0602 – Machine Learning

CO1 – AT2 – Concept Mapping

6. In the context of the Find-S algorithm, consider a dataset used to determine whether a person will go for a workout based on attributes such as Weather, Temperature, Time, and Mood. Apply the Find-S algorithm and determine the maximally specific hypothesis. Show the hypothesis updates after each positive example only. (5 Marks)

Dataset:

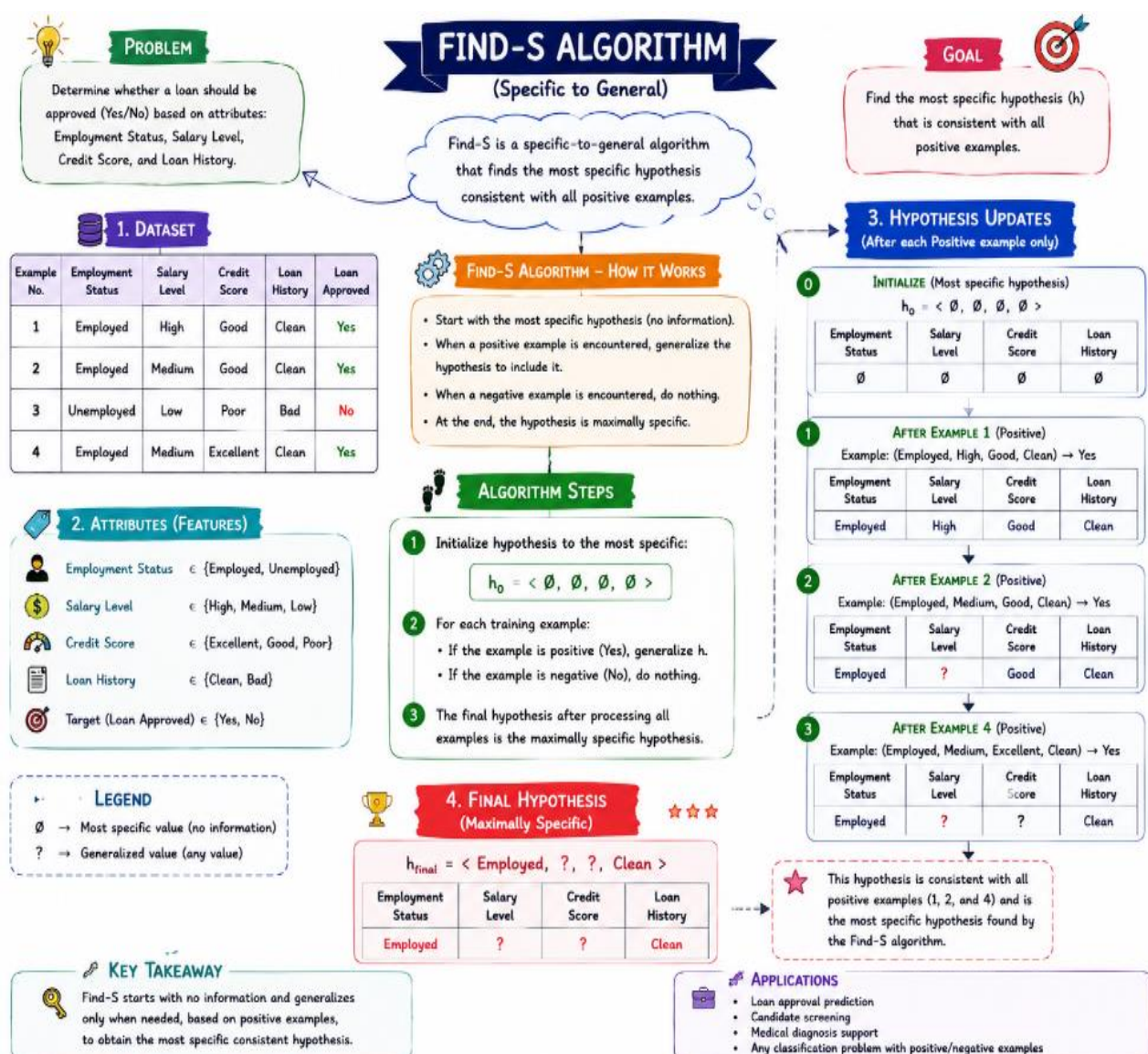
Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Evening	Happy	Yes



7. In the context of the Find-S algorithm, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples. (5 Marks)

Dataset:

Example	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

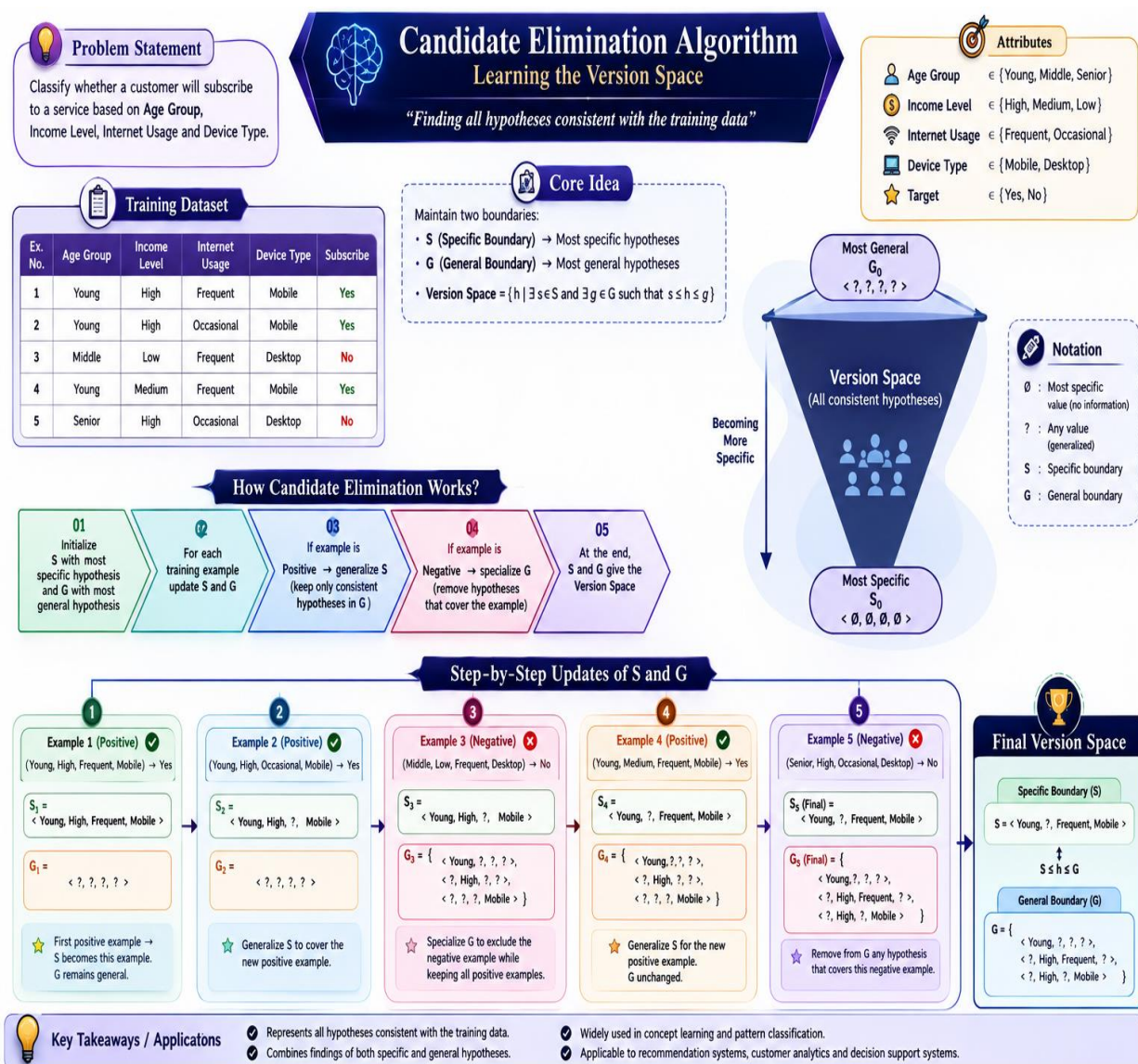


8. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age

Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the specific (S) and general (G) boundaries. Show step-by-step updates after each training example. (5 Marks)

Dataset:

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senoir	High	Occasional	Desktop	No

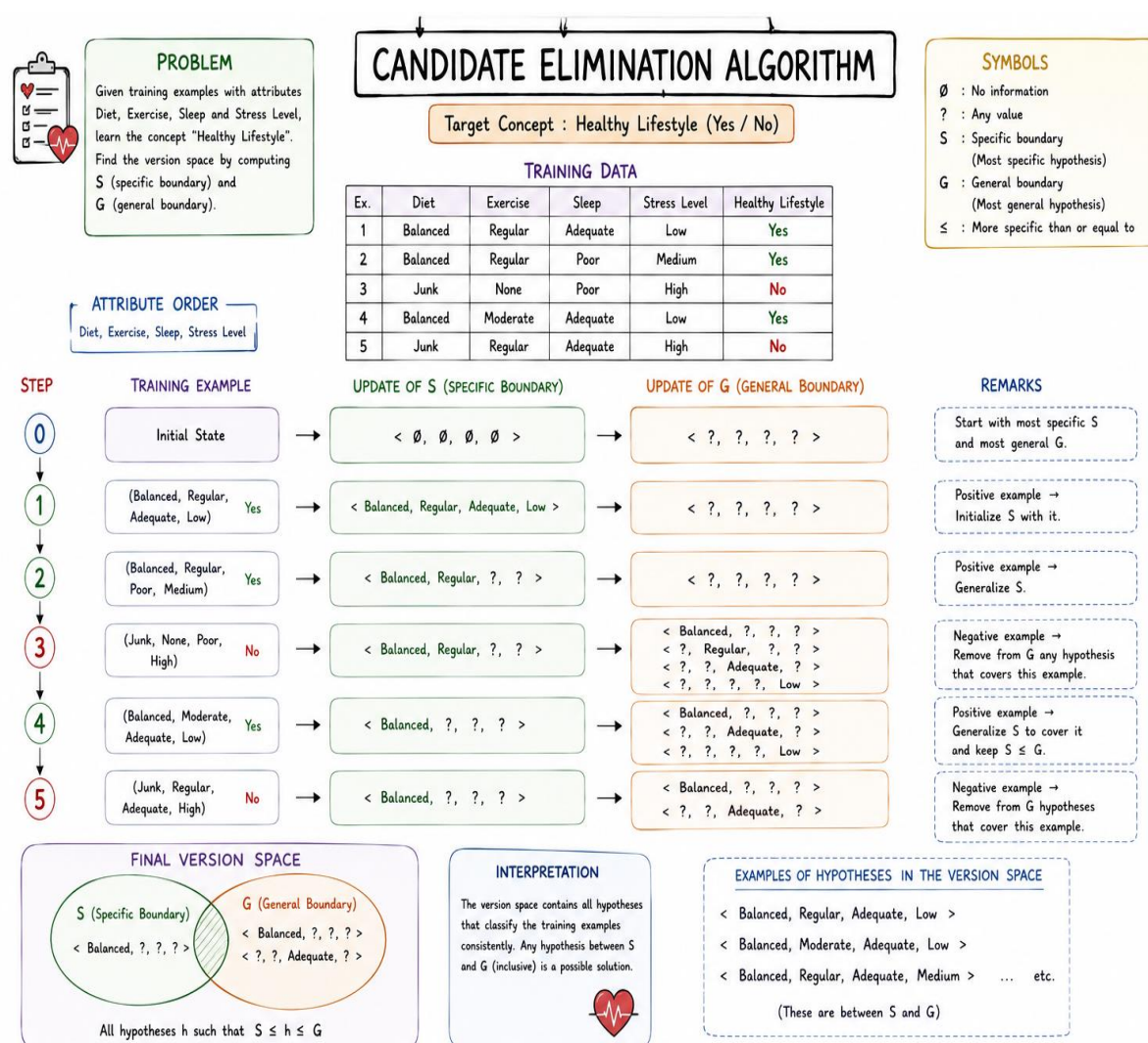


9. In the context of the Candidate Elimination algorithm, consider a dataset used to

classify whether a patient follows a healthy lifestyle based on attributes such as Diet, Exercise, Sleep, and Stress Level. Apply the algorithm to compute the version space (S and G boundaries) and show the updates after each example. (5 Marks)

Dataset:

Example	Diet	Exercise	Sleep	Stress Level	Healthy Lifestyle
1	Balanced	Regular	Adequate	Low	Yes
2	Balanced	Regular	Poor	Medium	Yes
3	Junk	None	Poor	High	No
4	Balanced	Moderate	Adequate	Low	Yes
5	Junk	Regular	Adequate	High	No



10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their

selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions.

